Figure 1. One platform, two cooperating modes, in a live energy OT environment

SOC Mode = deception-based active defence. Research Mode = offline analyst copilot. They share one data layer: live attacker behaviour informs research, research insight informs defence.

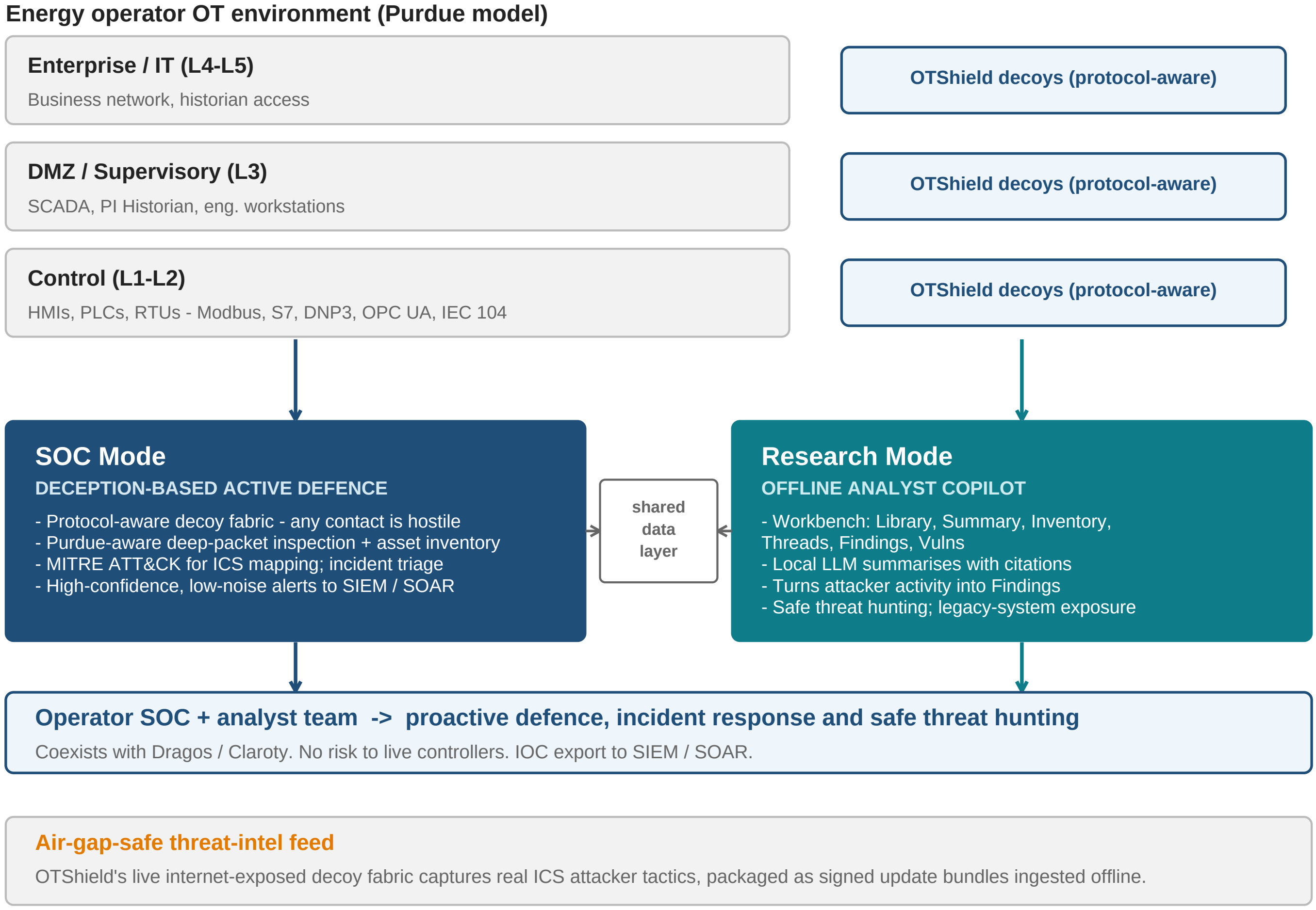

Table 1. Mapping to the four competition challenges

Challenge	Mode	How OTShield delivers it in this project
1. Proactive CNI defence & incident response	**SOC Mode**	High-confidence decoy detections + DPI; alerts to SIEM/SOAR; triage and response with the operator SOC team.
2. Detection & eviction of sophisticated actors	**SOC Mode**	Surfaces protocol-aware intent (unauthorised writes, interlock overrides, remote discovery) mapped across ATT&CK for ICS.
3. Threat hunting for CNI organisations	**Research Mode**	Copilot lets analysts query detections, manuals and asset data together and follow leads safely, with zero risk to live controllers.
4. Legacy system vulnerabilities in CNI	**Both modes**	SOC Mode inventories and risk-scores legacy assets; Research Mode copilot reasons over them to show how they could be reached and abused.

Figure 2. Readiness: TRL 7 today -> TRL 8 at project end

Today (TRL 7)
Both modes operational: decoy fabric validated against real ICS attackers (157 / 20 countries / 12,968 attacks); Research Mode copilot working offline.

Q1-Q2 (TRL 7)
Deploy both modes into the energy operator OT environment; tune to real assets, protocols and legacy equipment; first detection and hunting campaigns.

Q3 (TRL 7-8)
Continuous operation; incident-response and hunting exercises; capture detection-rate, time-to-detect and analyst-time-saved metrics; security review.

Q4 (TRL 8)
Qualified in operational environment; operator acceptance; commercialisation plan and development case study delivered.

Why it meets the challenge

One integrated platform - SOC Mode for deception-based active defence, Research Mode for the analyst copilot - is proven, accepted and qualified inside a real UK energy operator, delivering all four challenge areas and ready to scale across the sector.